

Networking Fundamentals

What is Network

A network is simply a collection of devices connected together to exchange data.

Whenever you send a message or open a website, your device communicates with another device somewhere in the world.

Examples:

- Mobile connected to Wi-Fi
- Computer connected to a printer
- Office computers connected together
- Internet connecting billions of devices"

Draw:

```
Laptop ---- Router ---- Internet
Mobile ----/
```

Types of Networks (15 Minutes)

LAN

"LAN stands for Local Area Network.

A LAN covers a small geographical area."

Examples:

- Home network
- Computer lab
- Office floor

Draw:

```
PC1 ---- Switch ---- PC2
          |
          PC3
```

WAN

"WAN stands for Wide Area Network.

A WAN connects multiple LANs over large distances."

Examples:

- Branch offices
- Banks
- Internet

Draw:

```
Office A ----- Internet ----- Office B
```

Client-Server Model

"When you open google.com, your computer requests information from Google's server."

Draw:

```
Client -----> Server  
Request
```

```
Client <----- Server  
Response
```

Examples:

Client	Server
Browser	Web Server
Outlook	Mail Server
Mobile App	Backend Server

Ask Students:

"Who is the client when you open YouTube?"

IP Address

"Every device on a network requires an identity."

This identity is called an IP Address."

```
192.168.0.10
```

Private IP

Used inside organizations.

Examples:

```
192.168.x.x  
10.x.x.x  
172.16.x.x - 172.31.x.x
```

192.168.0.0 - 192.168.255.255

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

Public IP

Assigned by ISP.

Visible on Internet.

- [whatismyip](#) - site

DNS

"If Google's IP is 142.250.xxx.xxx, do you remember it?"

"So we use names."

Example:

```
google.com
```

DNS converts:

```
google.com  
↓  
142.xxx.xxx.xxx
```

DNS = Internet Phonebook

```
User  
|  
DNS Query  
|  
DNS Server
```

```
|  
IP Address
```

Gateway

"Imagine your classroom has only one door.

Anyone leaving the room must use that door.

Similarly, a network has a gateway."

Draw:

```
PC  
|  
Gateway  
|  
Internet
```

Gateway is the exit point from a network.

DHCP (Dynamic Host Control Protocol)

DHCP automatically provides:

- IP Address
- Gateway
- DNS
- Subnet Mask

DORA

Process:

```
D: Discover  
O: Offer  
R: Request  
A: Acknowledge
```

NAT

OSI Model (Open Systems Interconnection)

7 Application	- HTTP, HTTPS, DNS, FTP, SMTP
6 Presentation	- SSL/TLS, JPEG, MP4, ASCII
5 Session	- NetBIOS, Cookies

- | | |
|-------------|------------------------------|
| 4 Transport | - TCP, UDP, Port Numbers |
| 3 Network | - IP, ICMP, Router |
| 2 Data Link | - ARP, MAC Address, Switch |
| 1 Physical | - Ethernet cable, Wi-Fi, Hub |

Remember: (In Reverse Order)

Please
Do
Not
Throw
Sausage
Pizza
Away

TCP vs UDP

TCP:

- Reliable - 3-way Handshak
- Acknowledgement
- Error Checking

Examples:

- HTTPS
- SSH

UDP:

- Faster
- No acknowledgement

Examples:

- DNS
- Streaming
- Gaming

Common Ports

Service	Port
HTTP	80
HTTPS	443
SMTP	25
MySQL	3306

Service	Port
SSH	22
FTP	20,21
DHCP	67,68
DNS	53
RDP	3389
POP3	110
IMAP	143
POP3S	993
IMAPS	995
Kerberos	88